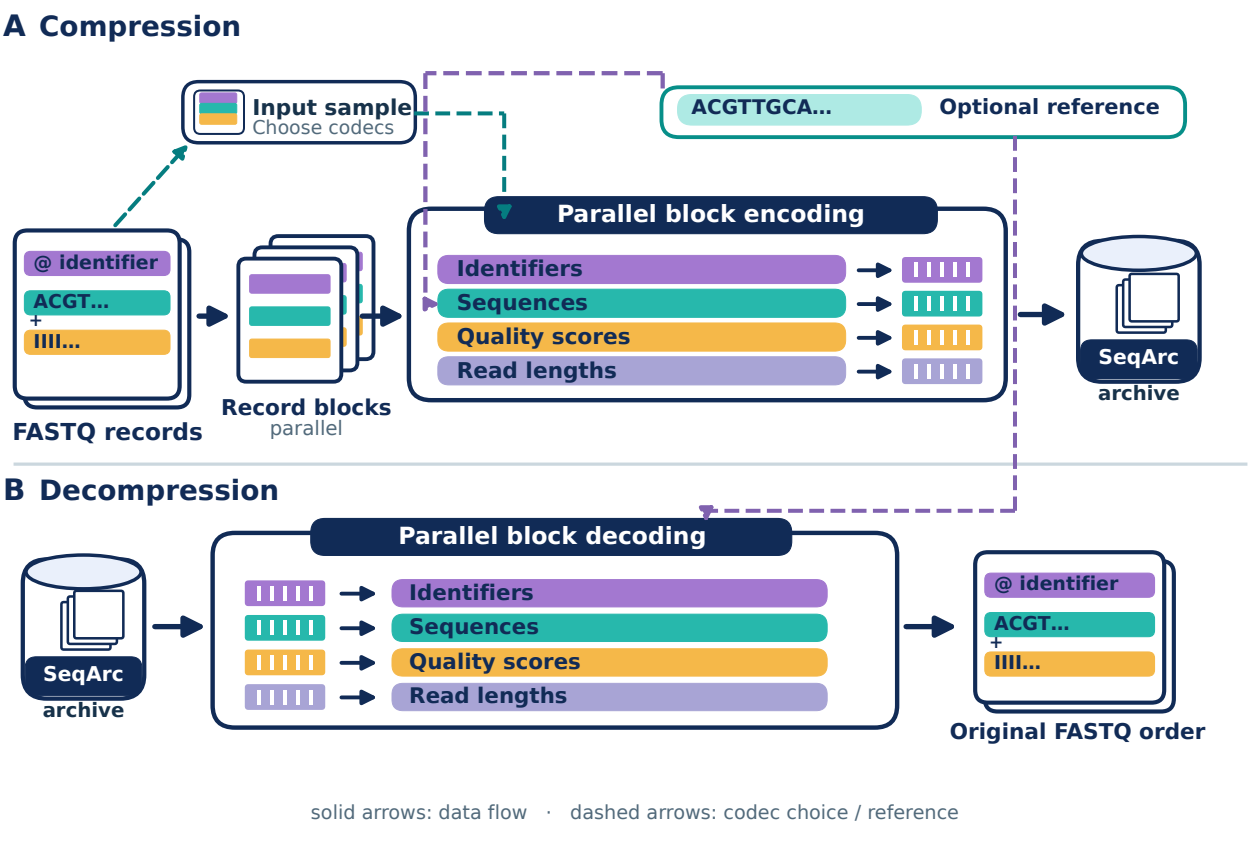
A Compression
Input sample
Choose codecs
ACGTTGCA…
Optional reference
@ identifier
ACGT…
+
IIII…
FASTQ records
Record blocks
parallel
Parallel block encoding
Identifiers
Sequences
Quality scores
Read lengths
SeqArc
archive
B Decompression
SeqArc
archive
Parallel block decoding
Identifiers
Sequences
Quality scores
Read lengths
@ identifier
ACGT…
+
IIII…
Original FASTQ order
solid arrows: data flow · dashed arrows: codec choice / reference